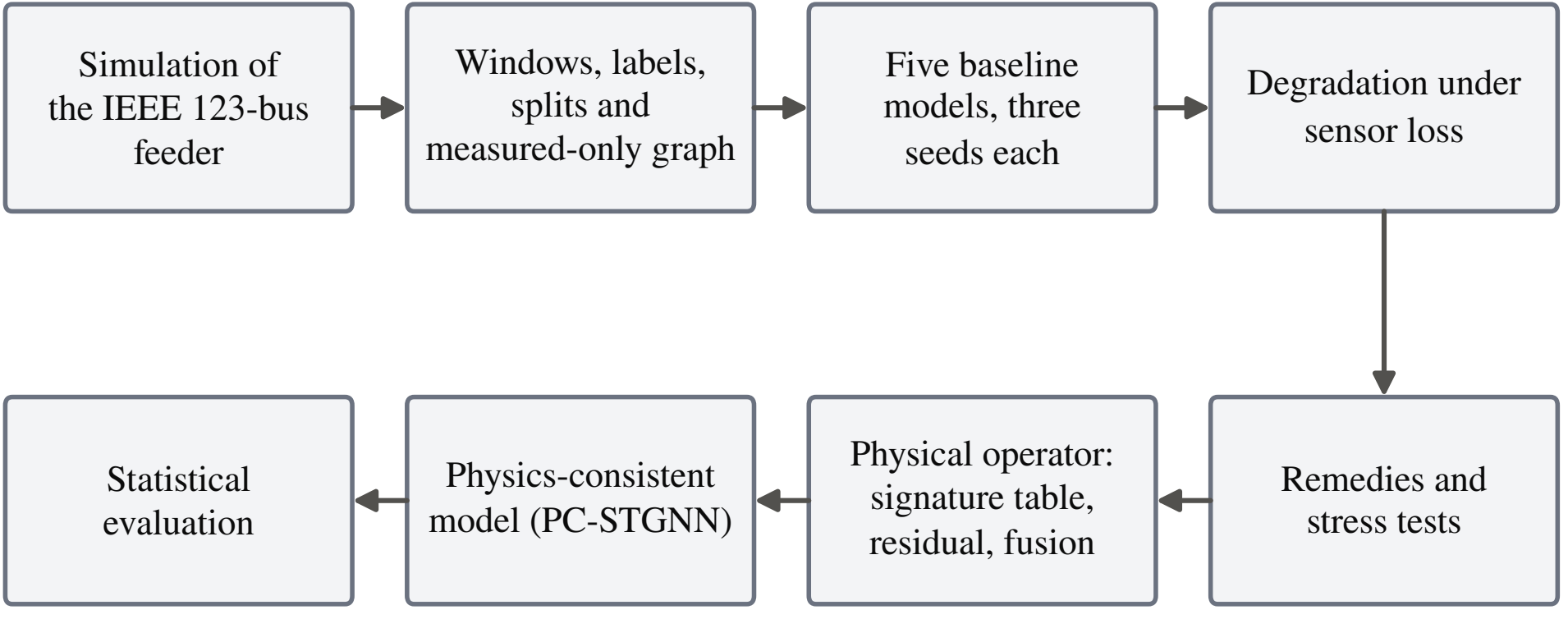
Simulation of the IEEE 123-bus feeder
Windows, labels, splits and measured-only graph
Five baseline models, three seeds each
Degradation under sensor loss
Remedies and stress tests
Physical operator: signature table, residual, fusion
Physics-consistent model (PC-STGNN)
Statistical evaluation